

# Supplementary Material: A formal knowledge base for metagenomics and geochemistry of the Rub' al Khali desert

## 1 Knowledge-representation patterns

This section specifies the knowledge-representation patterns that are not reproduced in the main article: sampling, protocol execution, taxonomic abundance, XRF geochemistry, the sequencing chain and sequencing quality control, and adds a worked example of the main article's laboratory-control pattern. Each pattern is given as its Description Logic axioms followed by a worked Turtle example taken from the released modules. Every printed triple and every prefixed IRI in this section is resolved against the staged modules by `scripts/validation/verify_manuscript_listings.py`, which fails if a listing drifts from the graph.

Listings use the prefix declarations of Table 1; prepend them to any listing to be executed against a local or deployed graph. The `rak_kb:` prefix covers the knowledge-base namespace without the `RAK_` local-name prefix, which is needed only for the PowerSoil extraction protocol `rak_kb:DNA_ext_powersoil`, the one individual in the released graph minted without it.

### 1.1 Laboratory-control worked example

Listing 1 instantiates the product-specification and evidence-bearing expected-taxon pattern for the whole-cell D6300 standard. The promoted specification-to-taxon statement supports direct queries, while the reified assertion preserves its subject, relation descriptor, taxon, status and evidence. Manufacturer genomic-DNA percentages remain in `metadata/controls/control_composition.tsv`; the RDF does not reinterpret them as expected 16S read fractions.

**Listing 1** Turtle representation of a D6300 composition specification and one expected-taxon assertion.

```
rak:CTL_24320eab3380ef6d405c590160b8d3310c2d641b7621bcef232f6d05cf4c2e60
a rak:0000301 ;
sio:000339
rak:CTL_99a452c18c4a63a400f55b2fe1abbf512ced2a0248d9952c25e441622268d409 .

rak:CTL_99a452c18c4a63a400f55b2fe1abbf512ced2a0248d9952c25e441622268d409
a rak:0000316 ;
sio:000338
rak:CTL_24320eab3380ef6d405c590160b8d3310c2d641b7621bcef232f6d05cf4c2e60 ;
```

**Table 1** Prefix declarations used in all Turtle and SPARQL listings.

| Prefix     | Namespace                                   |
|------------|---------------------------------------------|
| rak:       | https://rubalkhali.science/kb/RAK_          |
| rak_kb:    | https://rubalkhali.science/kb/              |
| sio:       | http://semanticscience.org/resource/SIO_    |
| owl:       | http://www.w3.org/2002/07/owl#              |
| pato:      | http://purl.obolibrary.org/obo/PATO_        |
| uo:        | http://purl.obolibrary.org/obo/UO_          |
| envo:      | http://purl.obolibrary.org/obo/ENVO_        |
| ncbitaxon: | http://purl.obolibrary.org/obo/NCBITaxon_   |
| obo:       | http://purl.obolibrary.org/obo/             |
| rdf:       | http://www.w3.org/1999/02/22-rdf-syntax-ns# |
| rdfs:      | http://www.w3.org/2000/01/rdf-schema#       |
| xsd:       | http://www.w3.org/2001/XMLSchema#           |
| geo:       | http://www.opengis.net/ont/geosparql#       |
| dcterms:   | http://purl.org/dc/terms/                   |

```

dcterms:source
  <https://www.zymoresearch.com/products/zymbiomics-microbial-community-
    standard> ;
rak:2000092 ncbi:taxon:1280 .

rak:CTL_1393f741397200195c65c9357ce358f08ed0b6368bdb5bfd379bf71b03f832bd
  a rak:0000317 ;
rak:2000093
  rak:CTL_99a452c18c4a63a400f55b2fe1abbf512ced2a0248d9952c25e441622268d409 ;
rak:2000094
  rak:CTL_56698f459e82fdb69be8ef81d785e212b8b59d6ec4f88b16dc300b69b1c461c3 ;
rak:2000095 ncbi:taxon:1280 ;
rak:2000090
  rak:CTL_d5aa029f71c6879da7029c52c4aaef883080caf632d3eaa14223a67d0a808d9d ;
sio:000772
  rak:CTL_998abc71455e01cd0175f49f1f32bc70113e47b047b072b30db1f4fe0de5fb13 .

rak:CTL_998abc71455e01cd0175f49f1f32bc70113e47b047b072b30db1f4fe0de5fb13
  sio:000773
  rak:CTL_1393f741397200195c65c9357ce358f08ed0b6368bdb5bfd379bf71b03f832bd .

```

## 1.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

$$\text{SamplingProcess} \sqsubseteq \text{Process} \quad (1)$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \quad (2)$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget.Site} \sqcap \exists \text{hasOutput.Sample} \quad (3)$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom.Site} \quad (4)$$

Listing 2 is generated from the Trip 1, Site 1 deep-soil record in the development graph.

**Listing 2** Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:7000001 a sio:001419 ; # Material sample collection
  sio:000059 rak:6000001 . # hasMember

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000244 rak:1000001 . # isDerivedFrom
```

Read in prose, an instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output. The collection explicitly contains `rak:6000001`, and this deep-soil sample is asserted to be derived from the same site. The redundancy between `hasTarget` on the process and `isDerivedFrom` on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

### 1.3 Experimental protocol pattern

We model experimental protocols as processes specified by a protocol entity. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO’s general *Process–hasInput/hasOutput–Object* idiom and adds a specification link to a reusable protocol description.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \quad (5)$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \quad (6)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy}.\text{Protocol} \quad (7)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput}.\text{MaterialEntity} \sqcap \exists \text{hasOutput}.\text{MaterialEntity} \quad (8)$$

Listing 3 demonstrates the formalization of a DNA extraction process.

**Listing 3** Turtle representation of a DNA extraction protocol.

```
rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:5900001 . # hasAttribute (Concentration Quality)
```

In words, the protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak_kb:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract

carries a concentration quality (`rak:5900001`, class `rak:0000043`) which is itself quantified through a separate measurement process following the pattern of the measurement pattern of the main article. The measurement value that quantifies it is the separate individual `rak:4900001` (class `rak:0000044`), which points to the quality with `sio:000215` and back from it with `sio:000216`. Value and quality are deliberately distinct individuals in every module; conflating them is the single most common way to break the traversal described in the Data Records section of the main article.

## 1.4 Taxonomic abundance pattern

Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. Here, we use the conservative mapping (Methods, taxonomy normalization) to assign either a validated NCBI class, an unambiguous retained project class, or a deterministic lineage-scoped contextual class as the target of each observation. Unreviewed GTDB candidates are not asserted. Listing 4 shows how an aggregated taxon–rank observation from one profile is represented in the knowledge graph.

For instance, the sequence analysis for a specific profile yields an absolute sequence count associated with an audited taxonomic class. We construct an abundance quality that is an attribute of both that taxon and the source sequence dataset, binding the count back to the biological workflow. Formally, the pattern instantiates SIO’s *Collection–hasMember–Thing* and reciprocal *MeasurementValue–isMeasurementValueOf–Quality* idioms:

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \quad (9)$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \quad (10)$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \quad (11)$$

$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \quad (12)$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AbundanceQuality} \quad (13)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \quad (14)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{hasMeasurementValue.AbundanceObservation} \quad (15)$$

**Listing 4** Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000339 rak:L000012 ; # 16S amplicon processing protocol
  sio:000230 rak:7791009 ; # hasInput (FASTQ Dataset, ERR16783308)
  sio:000229 rak:7740001 . # hasOutput (Taxonomic Dataset)

rak:7740001 a rak:0000074 ; # taxon absolute abundance dataset
  sio:000059 rak:V00000001 . # hasMember (Observation Value)

rak:V00000001 a rak:0000076 ; # Absolute abundance value
  sio:000215 rak:Q00000001 ; # isMeasurementValueOf
  rak:2000025 "Domain: Bacteria" ; # classifier lineage
  rak:2000026 "853814.0"^^xsd:double . # absolute read count
```

```

rak:Q00000001 a rak:0000078, pato:0000070 ; # Absolute Abundance Quality
sio:000011 ncbitaxon:2, rak:7791009 ; # taxon and FASTQ bearers
sio:000216 rak:V00000001 . # hasMeasurementValue

```

Verbalized, the bioinformatic-workflow instance `rak:P290001` consumes the FASTQ dataset of run `ERR16783308` (`rak:7791009`, profile `V10Dr1`) and produces the absolute taxonomic dataset `rak:7740001`; that dataset has as one of its members the measurement value `rak:V00000001`, which records an absolute read count of 853,814 for the domain-rank record `Domain: Bacteria`. Its reciprocal quality `rak:Q00000001` is an attribute of both *Bacteria* (`ncbitaxon:2`) and the source FASTQ dataset. Each of the fourteen datasets produced by one workflow instance holds a single rank at one of the two abundance bases, so a rank-restricted query never has to filter observations by lineage depth. One aggregated taxon–rank entry is therefore expanded into a four-node sub-graph that preserves both the classifier lineage and its sequence-dataset provenance. Contextual lineages use the same pattern with their audited project IRI.

## 1.5 XRF geochemistry pattern

We model every XRF session as a process with multiple concentration-value outputs and a workflow-specific protocol. A laboratory session has an archived soil specimen as input. The field instrument log, by contrast, identifies a test and site but not a physical specimen; the regenerated graph therefore links the session to the recorded sampling site and does not fabricate a collected specimen input. Field-XRF and laboratory-XRF records remain distinguishable even when they report the same analyte. Analyte cross-references to ChEBI [1] and PubChem [2] are maintained in a single mapping ledger. The fail-closed audit resolves every non-null identifier against pinned ChEBI v247 and a dated PubChem snapshot, checking identifier existence, label, formula, charge and atom-versus-oxide entity type. Of 93 instrument channels, 79 have a supported ChEBI cross-reference and 91 have a formula-matching PubChem cross-reference. A missing external identifier remains explicitly null rather than being filled by label similarity. These are `rdfs:seeAlso` annotations for the reporting channel, not assertions that a particular chemical species or mineral occurs in a sample. In particular, the instrument’s  $P_2O_5$  channel is an oxide-equivalent reporting basis; its ChEBI cross-reference has the same empirical ratio ( $P_4O_{10}$ ) and is not a molecular-presence claim. The instrument’s Light Elements (LE) output is an instrument pseudo-analyte or mixture, not an individual chemical entity, and is therefore not mapped to ChEBI or PubChem as a compound.

The shared output pattern specializes the measurement pattern (the measurement pattern of the main article); only the laboratory specialization requires a soil-sample input:

$$\text{XRFProcess} \sqsubseteq \text{MeasurementProcess} \sqcap \exists \text{hasOutput.Concentration} \quad (16)$$

$$\text{LabXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasInput.SoilSample} \quad (17)$$

$$\text{FieldXRFProcess} \sqsubseteq \text{XRFProcess} \sqcap \exists \text{hasTarget.SamplingSite} \quad (18)$$

$$\text{Concentration} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf}.\text{AnalyteQuality} \quad (19)$$

Listing 5 is generated from the aluminium output of laboratory-XRF process `rak:P300072` in the current module.

**Listing 5** Turtle representation of an XRF geochemistry measurement.

```
rak:P300072 a rak:0000025 ; # XRF Measuring Process
sio:000230 rak:6001732 ; # hasInput (sample V1Sr1)
sio:000339 rak:L_Surface_Lab ; # laboratory protocol
sio:000229 rak:4302121 . # hasOutput

rak:4302121 a rak:0000507 ; # Al concentration value
sio:000215 rak:5302121 ; # isMeasurementValueOf
rak:2000012 "0.78"^^xsd:double . # Concentration

rak:5302121 a rak:0000107 ; # Al concentration quality
sio:000011 rak:6001732 ; # isAttributeOf
sio:000216 rak:4302121 . # hasMeasurementValue
```

In words, this laboratory-XRF process consumes archived specimen `rak:6001732` (sample `V1Sr1`) and produces an aluminium concentration value. The value points to its aluminium quality using `sio:000215`; the quality points back using `sio:000216`. A dedicated scan of every XRF source artifact (the processed laboratory table, the field log and all raw instrument exports) (`scripts/xrf/audit_xrf_unit_evidence.py`) finds no unit declaration attached to any concentration column; the only unit token in the exports is PPM, and it belongs to the lower-limit-of-detection column. The generated module therefore carries no unit assertion for XRF concentrations. Readers should treat them as unitless instrument readings on a consistent scale within a workflow and must not use them for absolute cross-workflow comparisons. In the regenerated module, field sessions use `sio:000291` (*has target*) to identify the recorded site and retain the instrument test identifier; they have no constructed “field deep soil material” specimen input. This distinction also prevents a shared graph pattern from implying numerical interchangeability between workflows.

## 1.6 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process–hasInput/hasOutput–Object* steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern of the protocol pattern above.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \quad (20)$$

$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput}.\text{SoilSample} \sqcap \exists \text{hasOutput}.\text{DNAExtract} \quad (21)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \quad (22)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput}.\text{DNAExtract} \sqcap \exists \text{hasOutput}.\text{AmpliconLibrary} \quad (23)$$

$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \quad (24)$$

$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput.AmpliconLibrary} \sqcap \exists \text{hasOutput.FASTQDataset} \quad (25)$$

$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \quad (26)$$

$$\text{FASTQDataset} \sqsubseteq \forall \text{hasMember.SequenceRead} \quad (27)$$

Listing 6 instantiates the complete sequencing chain.

**Listing 6** Turtle representation of the sequencing workflow from soil sample T1Dr1 to ENA run ERR16061083.

```
rak:P200923 a sio:000994 ; # DNA Extraction Process
sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
sio:000230 rak:6000577 ; # hasInput (T1Dr1 Soil Sample)
sio:000229 rak:6800540 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000065 ; # Library Preparation Process
sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
sio:000230 rak:6800540 ; # hasInput (DNA Extract)
sio:000229 rak:6690001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000066 ; # Sequencing Process
sio:000230 rak:6690001 ; # hasInput (Amplicon Library)
sio:000229 rak:7790001 . # hasOutput (FASTQ Dataset)

rak:7790001 a rak:0000063 ; # FASTQ Dataset
rdfs:seeAlso <https://identifiers.org/insdc.run/ERR16061083> .
```

Read as a chain, the DNA extraction process `rak:P200923` consumes the soil sample `rak:6000577` and produces the DNA extract `rak:6800540`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6690001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7790001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput/hasOutput` pattern, so a single SPARQL traversal can recover the recorded provenance from any node in this chain.

## 1.7 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the generic measurement pattern of the measurement pattern of the main article. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \quad (28)$$

$$\text{SequencingQCProcess} \sqsubseteq \exists \text{hasInput.FASTQDataset} \sqcap \exists \text{hasOutput.QCMeasurementValue} \quad (29)$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.QCQuality} \quad (30)$$

$$\begin{aligned} \text{QCQuality} &\sqsubseteq \exists \text{isAttributeOf.} \\ &\quad \text{FASTQDataset} \end{aligned} \quad (31)$$

Listing 7 shows the corresponding quality-control pattern.

**Listing 7** Turtle representation of forward-read GC content for ENA run ERR16062319.

```
rak:P350001 a rak:0000200 ; # Sequencing QC Process
  sio:000230 rak:7790263 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4450001 . # hasOutput (QC Value)

rak:4450001 a rak:0000217 ; # Forward GC Content Value
  sio:000215 rak:5550001 ; # isMeasurementValueOf
  sio:000221 uo:0000187 ; # Percent
  rak:2000074 "58.0"^^xsd:double . # has GC content

rak:5550001 a rak:0000205 ; # Forward GC Content Quality
  sio:000011 rak:7790263 ; # isAttributeOf (FASTQ Dataset)
  sio:000216 rak:4450001 . # hasMeasurementValue
```

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7790263` as input and produces the measurement value `rak:4450001`, which records a GC content of 58.0 percent and is the measurement value of a forward-read GC-content quality `rak:5550001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

## 2 Supplementary Tables

Table 2 reports the curated field observations, and Table 3 reports the audited XRF channel cross-references.

**Table 2:** Site-specific field environmental observations across all campaigns. Values are generated from the immutable source sheets and the versioned correction ledger; a dash denotes an unrecorded or quarantined value. Altitude was not recorded in these field sheets.

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 1 (2023) | 1    | 17/03/2023 | 07:58 | 19.1498 N, 45.1665 E | 20.7       | 928           | 62.5   |
| Trip 1 (2023) | 2    | 17/03/2023 | 08:21 | 19.2035 N, 45.2448 E | 21.3       | 928           | 60.4   |
| Trip 1 (2023) | 3    | 17/03/2023 | 08:50 | 19.3011 N, 45.3844 E | 22.6       | 928           | 55.4   |
| Trip 1 (2023) | 4    | 17/03/2023 | 09:20 | 19.3796 N, 45.5518 E | 24         | 928           | 51.0   |
| Trip 1 (2023) | 5    | 17/03/2023 | 10:29 | 19.4072 N, 45.6595 E | 28.8       | 928           | 40.6   |
| Trip 1 (2023) | 6    | 17/03/2023 | 11:07 | 19.4057 N, 45.8161 E | 29         | 928           | 42.9   |
| Trip 1 (2023) | 7    | 17/03/2023 | 11:35 | 19.4070 N, 45.9539 E | 28.7       | 928           | 49.4   |
| Trip 1 (2023) | 8    | 17/03/2023 | 12:05 | 19.3932 N, 46.1107 E | 30.6       | 928           | 41.5   |
| Trip 1 (2023) | 9    | 17/03/2023 | 12:44 | 19.4112 N, 46.3027 E | 35.9       | 928           | 36.3   |
| Trip 1 (2023) | 10   | 17/03/2023 | 13:12 | 19.4205 N, 46.4608 E | 32.5       | 928           | 39.0   |
| Trip 1 (2023) | 11   | 17/03/2023 | 13:39 | 19.4444 N, 46.6356 E | 31.1       | 930           | 37.5   |
| Trip 1 (2023) | 12   | 17/03/2023 | 14:04 | 19.4779 N, 46.8344 E | 34.6       | 932           | 30.0   |
| Trip 1 (2023) | 13   | 17/03/2023 | 15:42 | 19.5293 N, 47.0533 E | 28.9       | 936           | 29.2   |
| Trip 1 (2023) | 14   | 17/03/2023 | 16:09 | 19.5813 N, 47.2920 E | 30.2       | 938           | 37.8   |
| Trip 1 (2023) | 15   | 17/03/2023 | 16:33 | 19.6137 N, 47.4759 E | 31.6       | 946           | 33.0   |
| Trip 1 (2023) | 16   | 17/03/2023 | 16:55 | 19.6445 N, 47.6261 E | 20.7       | 948           | 33.3   |
| Trip 1 (2023) | 17   | 17/03/2023 | 17:23 | 19.6765 N, 47.8273 E | 20.5       | 952           | 32.7   |
| Trip 1 (2023) | 18   | 18/03/2023 | 06:49 | 19.7325 N, 48.0469 E | 20         | 954           | 50.0   |
| Trip 1 (2023) | 19   | 18/03/2023 | 07:15 | 19.8123 N, 48.2405 E | 22.9       | 960           | 51.4   |
| Trip 1 (2023) | 20   | 18/03/2023 | 07:55 | 19.8693 N, 48.4246 E | 24.3       | 964           | 39.6   |
| Trip 1 (2023) | 21   | 18/03/2023 | 08:20 | 19.9011 N, 48.4865 E | 29.4       | 962           | 30.9   |
| Trip 1 (2023) | 22   | 18/03/2023 | 08:45 | 19.9787 N, 48.6394 E | 29         | —             | 25.9   |
| Trip 1 (2023) | 23   | 18/03/2023 | 09:25 | 20.0493 N, 48.7733 E | 26.8       | 966           | 25.2   |
| Trip 1 (2023) | 24   | 18/03/2023 | 09:48 | 20.0927 N, 48.8994 E | 29.5       | 966           | 25.9   |
| Trip 1 (2023) | 25   | 18/03/2023 | 10:15 | 20.1003 N, 48.9542 E | 27.8       | 968           | 27.6   |
| Trip 1 (2023) | 26   | 18/03/2023 | 10:40 | 20.1488 N, 49.0787 E | 31.7       | 968           | 23.1   |
| Trip 1 (2023) | 27   | 18/03/2023 | 11:18 | 20.2039 N, 49.1960 E | 31.5       | 970           | 23.8   |
| Trip 1 (2023) | 28   | 18/03/2023 | 11:47 | 20.2438 N, 49.3709 E | 31.9       | 970           | 24.9   |
| Trip 1 (2023) | 29   | 18/03/2023 | 12:15 | 20.2598 N, 49.5675 E | 31.7       | 968           | 25.7   |
| Trip 1 (2023) | 30   | 18/03/2023 | 14:39 | 20.2693 N, 49.6837 E | 34.1       | 964           | 22.4   |
| Trip 1 (2023) | 31   | 18/03/2023 | 15:11 | 20.2946 N, 49.8914 E | 33.8       | 964           | 22.3   |

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| Expedition    | Site       | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 1 (2023) | 32         | 18/03/2023 | 15:44 | 20.3107 N, 50.0930 E | 34         | 968           | 22.0   |
| Trip 1 (2023) | 33         | 18/03/2023 | 16:15 | 20.3175 N, 50.3031 E | 34.8       | 968           | 20.1   |
| Trip 1 (2023) | 34         | 18/03/2023 | 16:46 | 20.3444 N, 50.5154 E | 34.1       | 970           | 21.1   |
| Trip 1 (2023) | 35         | 18/03/2023 | 17:24 | 20.3718 N, 50.7300 E | 33.7       | 976           | 20.8   |
| Trip 1 (2023) | 36         | 19/03/2023 | 09:40 | 20.4232 N, 51.1448 E | 29.8       | 982           | 15.9   |
| Trip 1 (2023) | 37         | 19/03/2023 | 10:10 | 20.4542 N, 51.3449 E | 33.5       | 984           | 14.4   |
| Trip 1 (2023) | 38         | 19/03/2023 | 10:30 | 20.4610 N, 51.3702 E | 31.9       | 984           | 15.1   |
| Trip 1 (2023) | 39         | 19/03/2023 | 11:05 | 20.5102 N, 51.5974 E | 36.4       | 984           | 16.0   |
| Trip 1 (2023) | 40         | 19/03/2023 | 11:40 | 20.5472 N, 51.8256 E | 30.5       | 980           | 30.4   |
| Trip 1 (2023) | 41         | 19/03/2023 | 12:16 | 20.5398 N, 51.9461 E | 35.4       | 980           | 19.8   |
| Trip 1 (2023) | 42         | 19/03/2023 | 12:41 | 20.5552 N, 52.1240 E | 35.4       | 984           | 20.5   |
| Trip 1 (2023) | 43         | 19/03/2023 | 13:05 | 20.5676 N, 52.2241 E | 35.2       | 984           | 22.8   |
| Trip 1 (2023) | 44         | 19/03/2023 | 13:40 | 20.5678 N, 52.4104 E | 38.4       | 984           | 19.6   |
| Trip 1 (2023) | 45         | 19/03/2023 | 13:59 | 20.5635 N, 52.4678 E | 37.1       | 986           | 20.1   |
| Trip 1 (2023) | 46         | 19/03/2023 | 14:29 | 20.5537 N, 52.6692 E | 35.7       | 987           | 21.1   |
| Trip 1 (2023) | 47         | 19/03/2023 | 15:00 | 20.5418 N, 52.8688 E | 37.3       | 986           | 20.3   |
| Trip 1 (2023) | 48         | 19/03/2023 | 17:00 | 20.5686 N, 53.0940 E | 35.3       | 984           | 23.7   |
| Trip 1 (2023) | 49         | 19/03/2023 | 17:30 | 20.6447 N, 53.2879 E | 35         | 986           | 21.7   |
| Trip 1 (2023) | 50         | 20/03/2023 | 07:14 | 20.7139 N, 53.4123 E | 29.8       | 988           | 31.3   |
| Trip 1 (2023) | 51         | 20/03/2023 | 07:37 | 20.7987 N, 53.4870 E | 30.2       | 992           | 32.5   |
| Trip 1 (2023) | 52         | 20/03/2023 | 08:02 | 20.8278 N, 53.5784 E | 31.1       | 992           | 29.7   |
| Trip 1 (2023) | 53         | 20/03/2023 | 09:03 | 20.8515 N, 53.7579 E | 34.8       | 992           | 26.0   |
| Trip 1 (2023) | 54         | 20/03/2023 | 10:03 | 20.9202 N, 53.9622 E | 32.1       | 992           | 26.7   |
| Trip 1 (2023) | 55         | 20/03/2023 | 10:24 | 20.9023 N, 54.0024 E | 33.2       | 990           | 27.9   |
| Trip 1 (2023) | 56         | 20/03/2023 | 11:05 | 20.8216 N, 54.2239 E | 30         | 988           | 35.5   |
| Trip 1 (2023) | 57         | 20/03/2023 | 11:28 | 20.7350 N, 54.4176 E | 31.3       | 988           | 38.1   |
| Trip 1 (2023) | 58         | 20/03/2023 | 11:50 | 20.6654 N, 54.5933 E | 33.4       | 986           | 39.4   |
| Trip 1 (2023) | 59         | 20/03/2023 | 12:04 | 20.6563 N, 54.6359 E | 37.1       | 986           | 43.1   |
| Trip 1 (2023) | 60         | 20/03/2023 | 12:49 | 20.7064 N, 54.7321 E | 37.1       | 986           | 34.5   |
| Trip 1 (2023) | water well | 20/03/2023 | 13:16 | 20.7159 N, 54.7997 E | 35.6       | 984           | 31.4   |
| Trip 1 (2023) | road       | 20/03/2023 | 13:53 | 20.8258 N, 54.9889 E | 35.1       | 982           | 32.2   |
| Trip 1 (2023) | road       | 20/03/2023 | 14:20 | 20.9341 N, 55.1261 E | 36         | 982           | 26.3   |
| Trip 1 (2023) | camground  | 20/03/2023 | 14:45 | 21.0311 N, 55.1598 E | 37         | 984           | 26.0   |
| Trip 2 (2023) | 1          | 14/07/2023 | 06:10 | 19.1498 N, 45.1665 E | 34.5       | —             | —      |
| Trip 2 (2023) | 2          | 14/07/2023 | 07:12 | 19.2035 N, 45.2448 E | 36.7       | —             | —      |

Continued on next page

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 2 (2023) | 3    | 14/07/2023 | 07:16 | 19.3011 N, 45.3844 E | 37.0       | —             | —      |
| Trip 2 (2023) | 4    | 14/07/2023 | 07:36 | 19.3796 N, 45.5518 E | 38.6       | —             | —      |
| Trip 2 (2023) | 5    | 14/07/2023 | 07:43 | 19.4072 N, 45.6595 E | 39.2       | —             | —      |
| Trip 2 (2023) | 6    | 14/07/2023 | 08:58 | 19.4057 N, 45.8161 E | 41.2       | —             | —      |
| Trip 2 (2023) | 7    | 14/07/2023 | 09:20 | 19.4070 N, 45.9539 E | 41.3       | —             | —      |
| Trip 2 (2023) | 8    | 14/07/2023 | 10:20 | 19.3932 N, 46.1107 E | 41.9       | —             | —      |
| Trip 3 (2024) | 1    | 17/02/2024 | 06:54 | 19.1498 N, 45.1665 E | 21.9       | 898           | 55     |
| Trip 3 (2024) | 2    | 17/02/2024 | 07:21 | 19.2035 N, 45.2448 E | 26.6       | 904           | 40.9   |
| Trip 3 (2024) | 3    | 17/02/2024 | 08:05 | 19.3011 N, 45.3844 E | 28.3       | 914           | 38.5   |
| Trip 3 (2024) | 4    | 17/02/2024 | 08:40 | 19.3796 N, 45.5518 E | 30.2       | 922           | 37.5   |
| Trip 3 (2024) | 5    | 17/02/2024 | 09:51 | 19.4072 N, 45.6595 E | 35.2       | 926           | 19.5   |
| Trip 3 (2024) | 6    | 17/02/2024 | 10:45 | 19.4057 N, 45.8161 E | 37.1       | 942           | 17.6   |
| Trip 3 (2024) | 7    | 17/02/2024 | 11:28 | 19.4070 N, 45.9539 E | 37         | 926           | 19.6   |
| Trip 3 (2024) | 8    | 17/02/2024 | 11:50 | 19.3932 N, 46.1107 E | 39         | 922           | 19     |
| Trip 3 (2024) | 9    | 17/02/2024 | 12:31 | 19.4112 N, 46.3027 E | 36.4       | 924           | 21.5   |
| Trip 3 (2024) | 10   | 17/02/2024 | 13:22 | 19.4205 N, 46.4608 E | 37.1       | 926           | 18     |
| Trip 3 (2024) | 11   | 17/02/2024 | 13:46 | 19.4444 N, 46.6356 E | 41.2       | 930           | 21     |
| Trip 3 (2024) | 12   | 17/02/2024 | 14:04 | 19.4779 N, 46.8344 E | 39.9       | 932           | 15     |
| Trip 3 (2024) | 13   | 17/02/2024 | 15:20 | 19.5293 N, 47.0533 E | 40         | 940           | 13.5   |
| Trip 3 (2024) | 14   | 17/02/2024 | 15:50 | 19.5813 N, 47.2920 E | 41.2       | 940           | 14.9   |
| Trip 3 (2024) | 15   | 17/02/2024 | 16:13 | 19.6137 N, 47.4759 E | 38.8       | 944           | 13.5   |
| Trip 3 (2024) | 16   | 17/02/2024 | 16:34 | 19.6445 N, 47.6261 E | 39.4       | 948           | 11.9   |
| Trip 3 (2024) | 17   | 18/02/2024 | 08:42 | 19.6765 N, 47.8273 E | 23         | 956           | 27     |
| Trip 3 (2024) | 18   | 18/02/2024 | 09:54 | 19.7325 N, 48.0469 E | 29.2       | 960           | 18.7   |
| Trip 3 (2024) | 19   | 18/02/2024 | 10:25 | 19.8123 N, 48.2405 E | 27.4       | 962           | 21.2   |
| Trip 3 (2024) | 20   | 18/02/2024 | 11:05 | 19.8693 N, 48.4246 E | 32.9       | —             | 17.7   |
| Trip 3 (2024) | 21   | 18/02/2024 | 12:08 | 19.9011 N, 48.4865 E | 31.3       | 964           | 31.321 |
| Trip 3 (2024) | 22   | 18/02/2024 | 12:52 | 19.9787 N, 48.6394 E | 31.3       | 968           | 21.3   |
| Trip 3 (2024) | 23   | 18/02/2024 | 13:35 | 20.0493 N, 48.7733 E | 24.4       | 966           | 30.3   |
| Trip 3 (2024) | 24   | 18/02/2024 | 14:18 | 20.0927 N, 48.8994 E | 25.5       | 966           | 31.3   |
| Trip 3 (2024) | 25   | 18/02/2024 | 14:38 | 20.1003 N, 48.9542 E | 31         | 966           | 24.3   |
| Trip 3 (2024) | 26   | 18/02/2024 | 15:12 | 20.1488 N, 49.0787 E | 31.9       | 966           | 25.6   |
| Trip 3 (2024) | 27   | 18/02/2024 | 15:45 | 20.2039 N, 49.1960 E | 32.4       | 968           | 23.7   |
| Trip 3 (2024) | 28   | 18/02/2024 | 16:24 | 20.2438 N, 49.3709 E | 27.4       | 968           | 27.8   |
| Trip 3 (2024) | 29   | 18/02/2024 | 16:56 | 20.2598 N, 49.5675 E | 27.3       | 968           | 27     |

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| Expedition    | Site           | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|----------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 3 (2024) | 30             | 19/02/2024 | 08:10 | 20.2693 N, 49.6837 E | 36.8       | 976           | 20.2   |
| Trip 3 (2024) | 31             | 19/02/2024 | 08:43 | 20.2946 N, 49.8914 E | 23.1       | 978           | 29.3   |
| Trip 3 (2024) | 32             | 19/02/2024 | 09:20 | 20.3107 N, 50.0930 E | 26.7       | 982           | 23     |
| Trip 3 (2024) | 33             | 19/02/2024 | 10:07 | 20.3175 N, 50.3031 E | 26.6       | 984           | 20.9   |
| Trip 3 (2024) | 34             | 19/02/2024 | 10:52 | 20.3444 N, 50.5154 E | 29.9       | 982           | 17.7   |
| Trip 3 (2024) | 35             | 19/02/2024 | 11:30 | 20.3718 N, 50.7300 E | 28.7       | 984           | 17.9   |
| Trip 3 (2024) | 36             | 19/02/2024 | 13:58 | 20.4232 N, 51.1448 E | 28.2       | 982           | 16.1   |
| Trip 3 (2024) | 37             | 19/02/2024 | 14:52 | 20.4542 N, 51.3449 E | 29.1       | 984           | 15.5   |
| Trip 3 (2024) | 38             | 19/02/2024 | 15:01 | 20.4610 N, 51.3702 E | 29.3       | 984           | 15.1   |
| Trip 3 (2024) | 39             | 19/02/2024 | 15:56 | 20.5102 N, 51.5974 E | 28.8       | 984           | 16     |
| Trip 3 (2024) | 40             | 19/02/2024 | 16:42 | 20.5472 N, 51.8256 E | 28.2       | 986           | 17.3   |
| Trip 3 (2024) | 41             | 19/02/2024 | 17:14 | 20.5398 N, 51.9461 E | 29.4       | 988           | 16.2   |
| Trip 3 (2024) | 42             | 20/02/2024 | 08:15 | 20.5552 N, 52.1240 E | 24.6       | 990           | 34.2   |
| Trip 3 (2024) | 43             | 20/02/2024 | 08:38 | 20.5676 N, 52.2241 E | 27         | 992           | 27.8   |
| Trip 3 (2024) | 44             | 20/02/2024 | 09:22 | 20.5678 N, 52.4104 E | 29.1       | 994           | 22     |
| Trip 3 (2024) | 45             | 20/02/2024 | 09:44 | 20.5635 N, 52.4678 E | 32.4       | 996           | 18.3   |
| Trip 3 (2024) | 46             | 20/02/2024 | 10:44 | 20.5537 N, 52.6692 E | 32.3       | 994           | 17.4   |
| Trip 3 (2024) | 47             | 20/02/2024 | 11:24 | 20.5418 N, 52.8688 E | 33.2       | 994           | 17.2   |
| Trip 3 (2024) | 48             | 20/02/2024 | 12:54 | 20.5686 N, 53.0940 E | 49.4       | 994           | 9.5    |
| Trip 3 (2024) | 49             | 20/02/2024 | 14:21 | 20.6447 N, 53.2879 E | 34         | 994           | 13.4   |
| Trip 3 (2024) | 50             | 20/02/2024 | 14:51 | 20.7139 N, 53.4123 E | 31.5       | 996           | 21.6   |
| Trip 3 (2024) | 51             | 20/02/2024 | 16:11 | 20.7987 N, 53.4870 E | 35.9       | 994           | 16.5   |
| Trip 3 (2024) | 52             | 20/02/2024 | 06:57 | 20.8278 N, 53.5784 E | 34.3       | 994           | 17.5   |
| Trip 3 (2024) | 53             | 21/02/2024 | 06:06 | 20.8515 N, 53.7579 E | 21.9       | 990           | 35     |
| Trip 3 (2024) | 54             | 21/02/2024 | 15:58 | 20.9202 N, 53.9622 E | 32.2       | 998           | 23.5   |
| Trip 3 (2024) | 55             | 21/02/2024 | 15:41 | 20.9023 N, 54.0024 E | 31.3       | 998           | 14.3   |
| Trip 3 (2024) | 56             | 21/02/2024 | 15:10 | 20.8216 N, 54.2239 E | 31.6       | 998           | 13.5   |
| Trip 3 (2024) | 57             | 21/02/2024 | 14:48 | 20.7350 N, 54.4176 E | 34.2       | 996           | 16.3   |
| Trip 3 (2024) | 58             | 21/02/2024 | 14:29 | 20.6654 N, 54.5933 E | 35.6       | 992           | 16.4   |
| Trip 3 (2024) | 59             | 21/02/2024 | 14:15 | 20.6563 N, 54.6359 E | 36.8       | 992           | 14.3   |
| Trip 3 (2024) | 60             | 21/02/2024 | 13:20 | 20.7064 N, 54.7321 E | 37.1       | 992           | 14.3   |
| Trip 3 (2024) | oill spill     | 17/02/2024 | 10:25 | 19.4054 N, 45.7271 E | 35.1       | —             | 20     |
| Trip 3 (2024) | water well     | 17/02/2024 | 14:23 | 19.5014 N, 46.9015 E | 39.2       | —             | 15.9   |
| Trip 3 (2024) | Camell         | 17/02/2024 | 16:49 | 19.6540 N, 47.6883 E | 37.6       | —             | 25     |
| Trip 3 (2024) | plant gradient | 18/02/2024 | 09:25 | 19.7092 N, 47.9593 E | 25         | 958           | 24     |

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| Expedition                      | Site                     | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------------------------|--------------------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 3 (2024)                   | hot spring               | 19/02/2024 | 12:25 | 20.3972 N, 50.9655 E | 32.1       | 984           | 14.8   |
| Trip 1 auxiliary/revisit (2023) | 12.5+A                   | 17/03/2023 | 14:22 | 19.5015 N, 46.9014 E | 33         | 934           | 30     |
| Trip 1 auxiliary/revisit (2023) | 19.5                     | 18/03/2023 | 07:20 | 19.8193 N, 48.2402 E | 21.8       | 960           | 61.9   |
| Trip 1 auxiliary/revisit (2023) | 27+                      | 18/03/2023 | 11:21 | 20.2039 N, 49.1960 E | 31.5       | 970           | 23.8   |
| Trip 1 auxiliary/revisit (2023) | 31urine                  | 18/03/2023 | 15:14 | 20.2946 N, 49.8914 E | 33.8       | 964           | 22.3   |
| Trip 1 auxiliary/revisit (2023) | 32+                      | 18/03/2023 | 15:44 | 20.3107 N, 50.0930 E | 34         | 968           | 22     |
| Trip 1 auxiliary/revisit (2023) | 41+                      | 19/03/2023 | 12:16 | 20.5398 N, 51.9461 E | 35.4       | 980           | 19.8   |
| Trip 1 auxiliary/revisit (2023) | 47+                      | 19/03/2023 | 15:00 | 20.5418 N, 52.8688 E | 37.3       | 986           | 20.3   |
| Trip 1 auxiliary/revisit (2023) | 47+grass                 | 19/03/2023 | 15:00 | 20.5418 N, 52.8688 E | 37.3       | 986           | 20.3   |
| Trip 1 auxiliary/revisit (2023) | 48+                      | 19/03/2023 | 17:00 | 20.5686 N, 53.0940 E | 35.3       | 984           | 23.7   |
| Trip 1 auxiliary/revisit (2023) | 48+                      | 19/03/2023 | 17:00 | 20.5686 N, 53.0940 E | 35.3       | 984           | 23.7   |
| Trip 1 auxiliary/revisit (2023) | 52AD                     | 20/03/2023 | 08:02 | 20.8278 N, 53.5784 E | 31.1       | 992           | 29.7   |
| Trip 1 auxiliary/revisit (2023) | 54F                      | 20/03/2023 | 10:16 | 20.9202 N, 53.9622 E | 34.8       | 990           | 26.9   |
| Trip 1 auxiliary/revisit (2023) | 35 + WaterWellRestStop   | 18/03/2023 | 18:48 | 20.3972 N, 50.9657 E | —          | 976           | —      |
| Trip 1 auxiliary/revisit (2023) | 35 + hot spring filter1  | 19/03/2023 | 08:03 | 20.3972 N, 50.9657 E | 28.3       | 980           | 17.7   |
| Trip 1 auxiliary/revisit (2023) | 35 + hot spring filter 2 | 19/03/2023 | 08:11 | 20.3972 N, 50.9657 E | 28.3       | 980           | 17.7   |
| Trip 4 (2024)                   | 1                        | 29/08/2024 | 06:04 | 19.1498 N, 45.1665 E | 30.83      | —             | 19.4   |
| Trip 4 (2024)                   | 2                        | 29/08/2024 | 06:24 | 19.2035 N, 45.2448 E | 31.5       | —             | 18.4   |
| Trip 4 (2024)                   | 3                        | 29/08/2024 | 06:53 | 19.3011 N, 45.3844 E | 33.61      | —             | 16.5   |
| Trip 4 (2024)                   | 4                        | 29/08/2024 | 07:15 | 19.3796 N, 45.5518 E | 35.11      | —             | 15.5   |
| Trip 4 (2024)                   | 5                        | 29/08/2024 | 07:29 | 19.4072 N, 45.6595 E | 36.67      | —             | 14     |
| Trip 4 (2024)                   | 6                        | 29/08/2024 | 08:07 | 19.4057 N, 45.8161 E | 37.06      | —             | 13.3   |
| Trip 4 (2024)                   | 7                        | 29/08/2024 | 08:32 | 19.4070 N, 45.9539 E | 37.56      | —             | 13     |
| Trip 4 (2024)                   | 8                        | 29/08/2024 | 08:57 | 19.3932 N, 46.1107 E | 37.39      | —             | 14.2   |
| Trip 4 (2024)                   | 9                        | 29/08/2024 | 09:28 | 19.4112 N, 46.3027 E | 39         | —             | 13.5   |
| Trip 4 (2024)                   | 10                       | 29/08/2024 | 09:50 | 19.4205 N, 46.4608 E | 39.56      | —             | 13.4   |
| Trip 4 (2024)                   | 11                       | 29/08/2024 | 10:14 | 19.4444 N, 46.6356 E | 42         | —             | 12.4   |
| Trip 4 (2024)                   | 12                       | 29/08/2024 | 10:35 | 19.4779 N, 46.8344 E | 42.94      | —             | 21.5   |
| Trip 4 (2024)                   | 13                       | 29/08/2024 | 11:30 | 19.5293 N, 47.0533 E | 41.94      | —             | 16.5   |
| Trip 4 (2024)                   | 14                       | 29/08/2024 | 11:50 | 19.5813 N, 47.2920 E | 42.56      | —             | 18     |
| Trip 4 (2024)                   | 15                       | 29/08/2024 | 12:06 | 19.6137 N, 47.4759 E | 44.11      | —             | 15.2   |
| Trip 4 (2024)                   | 16                       | 29/08/2024 | 12:21 | 19.6445 N, 47.6261 E | 43.5       | —             | 15.7   |
| Trip 4 (2024)                   | 17                       | 29/08/2024 | 12:49 | 19.6765 N, 47.8273 E | 43.11      | —             | 15.2   |
| Trip 4 (2024)                   | 18                       | 29/08/2024 | 13:24 | 19.7325 N, 48.0469 E | 43.83      | —             | 12.8   |
| Trip 4 (2024)                   | 19                       | 29/08/2024 | 13:52 | 19.8123 N, 48.2405 E | 42.83      | —             | 14.1   |

Continued on next page

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 4 (2024) | 20   | 29/08/2024 | 13:57 | 19.8693 N, 48.4246 E | 41.94      | —             | 14.5   |
| Trip 4 (2024) | 21   | 29/08/2024 | 14:32 | 19.9011 N, 48.4865 E | 41.72      | —             | 13.1   |
| Trip 4 (2024) | 22   | 29/08/2024 | 14:50 | 19.9787 N, 48.6394 E | 43.33      | —             | 13     |
| Trip 4 (2024) | 23   | 29/08/2024 | 15:21 | 20.0493 N, 48.7733 E | 44.28      | —             | 12     |
| Trip 4 (2024) | 24   | 29/08/2024 | 15:55 | 20.0927 N, 48.8994 E | 43.17      | —             | 12.7   |
| Trip 4 (2024) | 25   | 29/08/2024 | 16:45 | 20.1003 N, 48.9542 E | 43.5       | —             | 11.6   |
| Trip 4 (2024) | 26   | 29/08/2024 | 17:02 | 20.1488 N, 49.0787 E | 42.83      | —             | 12.3   |
| Trip 4 (2024) | 27   | 29/08/2024 | 17:28 | 20.2039 N, 49.1960 E | 41.83      | —             | 12.6   |
| Trip 4 (2024) | 28   | 29/08/2024 | 17:56 | 20.2438 N, 49.3709 E | 40.83      | —             | 13.3   |
| Trip 4 (2024) | 29   | 29/08/2024 | 18:21 | 20.2598 N, 49.5675 E | 40.78      | —             | 13.3   |
| Trip 4 (2024) | 30   | 30/08/2024 | 05:51 | 20.2693 N, 49.6837 E | 36.5       | —             | 27.7   |
| Trip 4 (2024) | 31   | 30/08/2024 | 06:08 | 20.2946 N, 49.8914 E | 35.89      | —             | 27.3   |
| Trip 4 (2024) | 32   | 30/08/2024 | 06:30 | 20.3107 N, 50.0930 E | 35.72      | —             | 21.9   |
| Trip 4 (2024) | 33   | 30/08/2024 | 06:50 | 20.3175 N, 50.3031 E | 35.89      | —             | 23     |
| Trip 4 (2024) | 34   | 30/08/2024 | 07:11 | 20.3444 N, 50.5154 E | 36.39      | —             | 23.5   |
| Trip 4 (2024) | 35   | 30/08/2024 | 07:25 | 20.3718 N, 50.7300 E | 38.5       | —             | 21.2   |
| Trip 4 (2024) | 36   | 30/08/2024 | 08:01 | 20.4232 N, 51.1448 E | 39.28      | —             | 31.8   |
| Trip 4 (2024) | 37   | 30/08/2024 | 08:47 | 20.4542 N, 51.3449 E | 39.44      | —             | 23.2   |
| Trip 4 (2024) | 38   | 30/08/2024 | 09:14 | 20.4610 N, 51.3702 E | 41.06      | —             | 20     |
| Trip 4 (2024) | 39   | 30/08/2024 | 09:20 | 20.5102 N, 51.5974 E | 42.44      | —             | 19.5   |
| Trip 4 (2024) | 40   | 30/08/2024 | 09:49 | 20.5472 N, 51.8256 E | 42.06      | —             | 19     |
| Trip 4 (2024) | 41   | 30/08/2024 | 10:25 | 20.5398 N, 51.9461 E | 43.17      | —             | 17     |
| Trip 4 (2024) | 42   | 30/08/2024 | 10:50 | 20.5552 N, 52.1240 E | 43.67      | —             | 16.2   |
| Trip 4 (2024) | 43   | 30/08/2024 | 11:19 | 20.5676 N, 52.2241 E | 42.78      | —             | 17     |
| Trip 4 (2024) | 44   | 30/08/2024 | 11:40 | 20.5678 N, 52.4104 E | 41.94      | —             | 18     |
| Trip 4 (2024) | 45   | 30/08/2024 | 12:12 | 20.5635 N, 52.4678 E | 40.94      | —             | 17.1   |
| Trip 4 (2024) | 46   | 30/08/2024 | 12:22 | 20.5537 N, 52.6692 E | 44         | —             | 15.7   |
| Trip 4 (2024) | 47   | 30/08/2024 | 13:16 | 20.5418 N, 52.8688 E | 42.94      | —             | 15.3   |
| Trip 4 (2024) | 48   | 30/08/2024 | 13:57 | 20.5686 N, 53.0940 E | 42.5       | —             | 18.2   |
| Trip 4 (2024) | 49   | 30/08/2024 | 14:28 | 20.6447 N, 53.2879 E | 44         | —             | 14.4   |
| Trip 4 (2024) | 50   | 31/08/2024 | 06:08 | 20.7139 N, 53.4123 E | 35.61      | —             | 25.1   |
| Trip 4 (2024) | 51   | 31/08/2024 | 06:34 | 20.7987 N, 53.4870 E | 36.11      | —             | 25     |
| Trip 4 (2024) | 52   | 31/08/2024 | 06:54 | 20.8278 N, 53.5784 E | 36.28      | —             | 25.5   |
| Trip 4 (2024) | 53   | 31/08/2024 | 09:00 | 20.8515 N, 53.7579 E | 42.61      | —             | 17.5   |
| Trip 4 (2024) | 54   | 31/08/2024 | 09:34 | 20.9202 N, 53.9622 E | 42.5       | —             | 17.5   |

Continued on next page

| Expedition    | Site | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------|------------|-------|----------------------|------------|---------------|--------|
| Trip 4 (2024) | 55   | 31/08/2024 | 09:48 | 20.9023 N, 54.0024 E | 40.72      | —             | 20     |
| Trip 4 (2024) | 56   | 31/08/2024 | 10:16 | 20.8216 N, 54.2239 E | 40.44      | —             | 21.4   |
| Trip 4 (2024) | 57   | 31/08/2024 | 10:40 | 20.7350 N, 54.4176 E | 42.78      | —             | 18     |
| Trip 4 (2024) | 58   | 31/08/2024 | 10:58 | 20.6654 N, 54.5933 E | 43.5       | —             | 15     |
| Trip 4 (2024) | 59   | 31/08/2024 | 11:06 | 20.6563 N, 54.6359 E | 43         | —             | 14.9   |
| Trip 4 (2024) | 60   | 31/08/2024 | 11:28 | 20.7064 N, 54.7321 E | 42.89      | —             | 18.1   |
| Trip 5 (2025) | 1    | 12/10/2025 | 07:32 | 19.1498 N, 45.1665 E | 26.9       | 894           | 24.3   |
| Trip 5 (2025) | 2    | 12/10/2025 | 08:05 | 19.2035 N, 45.2448 E | 27.8       | 902           | 23.4   |
| Trip 5 (2025) | 3    | 12/10/2025 | 08:38 | 19.3011 N, 45.3844 E | 29         | 912           | 23     |
| Trip 5 (2025) | 4    | 12/10/2025 | 09:20 | 19.3796 N, 45.5518 E | 32.4       | 922           | 18.8   |
| Trip 5 (2025) | 5    | 12/10/2025 | 10:00 | 19.4072 N, 45.6595 E | 33.4       | 924           | 19.2   |
| Trip 5 (2025) | 6    | 12/10/2025 | 10:50 | 19.4057 N, 45.8161 E | 36.7       | 916           | 14.1   |
| Trip 5 (2025) | 7    | 12/10/2025 | 11:18 | 19.4070 N, 45.9539 E | 35         | 918           | 13.4   |
| Trip 5 (2025) | 8    | 12/10/2025 | 11:55 | 19.3932 N, 46.1107 E | 34         | 918           | 12.9   |
| Trip 5 (2025) | 9    | 12/10/2025 | 12:33 | 19.4112 N, 46.3027 E | 35.5       | 918           | 12.5   |
| Trip 5 (2025) | 10   | 12/10/2025 | 12:58 | 19.4205 N, 46.4608 E | 35.6       | 920           | 10.5   |
| Trip 5 (2025) | 11   | 12/10/2025 | 13:30 | 19.4444 N, 46.6356 E | 35.6       | 926           | 10.7   |
| Trip 5 (2025) | 12   | 12/10/2025 | 13:57 | 19.4779 N, 46.8344 E | 36.3       | 928           | 10.3   |
| Trip 5 (2025) | 13   | 12/10/2025 | 15:25 | 19.5293 N, 47.0533 E | 36.5       | 934           | 10.1   |
| Trip 5 (2025) | 14   | 12/10/2025 | 15:51 | 19.5813 N, 47.2920 E | 35         | 936           | 11.9   |
| Trip 5 (2025) | 15   | 12/10/2025 | 16:22 | 19.6137 N, 47.4759 E | 34.7       | 940           | 11.9   |
| Trip 5 (2025) | 16   | 12/10/2025 | 16:42 | 19.6445 N, 47.6261 E | 33.9       | 942           | 13.2   |
| Trip 5 (2025) | 17   | 13/10/2025 | 07:05 | 19.6765 N, 47.8273 E | 26.4       | 948           | 32.1   |
| Trip 5 (2025) | 18   | 13/10/2025 | 07:42 | 19.7325 N, 48.0469 E | 26.6       | 950           | 27     |
| Trip 5 (2025) | 19   | 13/10/2025 | 08:15 | 19.8123 N, 48.2405 E | 28.2       | 954           | 27.5   |
| Trip 5 (2025) | 20   | 13/10/2025 | 09:04 | 19.8693 N, 48.4246 E | 32.9       | 958           | 23.5   |
| Trip 5 (2025) | 21   | 13/10/2025 | 09:30 | 19.9011 N, 48.4865 E | 33         | 958           | 25     |
| Trip 5 (2025) | 22   | 13/10/2025 | 10:10 | 19.9787 N, 48.6394 E | 35.5       | 962           | 20.8   |
| Trip 5 (2025) | 23   | 13/10/2025 | 10:56 | 20.0493 N, 48.7733 E | 36.1       | 960           | 21.6   |
| Trip 5 (2025) | 24   | 13/10/2025 | 11:39 | 20.0927 N, 48.8994 E | 36.3       | 958           | 18.2   |
| Trip 5 (2025) | 25   | 13/10/2025 | 11:55 | 20.1003 N, 48.9542 E | 35         | 960           | 18.4   |
| Trip 5 (2025) | 26   | 13/10/2025 | 12:16 | 20.1488 N, 49.0787 E | 34.3       | 960           | 19.5   |
| Trip 5 (2025) | 27   | 13/10/2025 | 12:42 | 20.2039 N, 49.1960 E | 36.4       | 960           | 16.6   |
| Trip 5 (2025) | 28   | 13/10/2025 | 13:08 | 20.2438 N, 49.3709 E | 38.8       | 960           | 15.5   |
| Trip 5 (2025) | 29   | 13/10/2025 | 13:56 | 20.2598 N, 49.5675 E | 37.5       | 966           | 14.9   |

Continued on next page

| Expedition    | Site       | Date       | Time  | Coordinates          | Temp. (°C) | Press. (mbar) | RH (%) |
|---------------|------------|------------|-------|----------------------|------------|---------------|--------|
| Trip 5 (2025) | 30         | 13/10/2025 | 14:13 | 20.2693 N, 49.6837 E | 37.5       | 966           | 14.2   |
| Trip 5 (2025) | 31         | 13/10/2025 | 14:46 | 20.2946 N, 49.8914 E | 35.9       | 968           | 17.5   |
| Trip 5 (2025) | 32         | 13/10/2025 | 15:38 | 20.3107 N, 50.0930 E | 39         | 970           | 12.8   |
| Trip 5 (2025) | 33         | 13/10/2025 | 16:10 | 20.3175 N, 50.3031 E | 37.9       | 972           | 11.9   |
| Trip 5 (2025) | 34         | 13/10/2025 | 16:58 | 20.3444 N, 50.5154 E | 34.3       | 974           | 17.9   |
| Trip 5 (2025) | 35         | 14/10/2025 | 06:58 | 20.3718 N, 50.7300 E | 35.1       | 982           | 27.1   |
| Trip 5 (2025) | 36         | 14/10/2025 | 08:36 | 20.4232 N, 51.1448 E | 33.2       | 982           | 22.6   |
| Trip 5 (2025) | 37         | 14/10/2025 | 09:01 | 20.4542 N, 51.3449 E | 32.3       | 984           | 23.7   |
| Trip 5 (2025) | 38         | 14/10/2025 | 09:13 | 20.4610 N, 51.3702 E | 33.2       | 984           | 22.3   |
| Trip 5 (2025) | 39         | 14/10/2025 | 09:50 | 20.5102 N, 51.5974 E | 37.2       | 986           | 22.1   |
| Trip 5 (2025) | 40         | 14/10/2025 | 10:40 | 20.5472 N, 51.8256 E | 39         | 984           | —      |
| Trip 5 (2025) | 41         | 14/10/2025 | 11:04 | 20.5398 N, 51.9461 E | 37.1       | 986           | 20.8   |
| Trip 5 (2025) | 42         | 14/10/2025 | 11:40 | 20.5552 N, 52.1240 E | 36         | 984           | 22.6   |
| Trip 5 (2025) | 43         | 14/10/2025 | 12:07 | 20.5676 N, 52.2241 E | 40.4       | 984           | 19.7   |
| Trip 5 (2025) | 44         | 14/10/2025 | 13:03 | 20.5678 N, 52.4104 E | 37.1       | 984           | 19.3   |
| Trip 5 (2025) | 45         | 14/10/2025 | 13:16 | 20.5635 N, 52.4678 E | 39.5       | 984           | 17.6   |
| Trip 5 (2025) | 46         | 14/10/2025 | 14:02 | 20.5537 N, 52.6692 E | 38.6       | 982           | 17.2   |
| Trip 5 (2025) | 47         | 14/10/2025 | 14:41 | 20.5418 N, 52.8688 E | 36.2       | 984           | 23     |
| Trip 5 (2025) | 48         | 14/10/2025 | 15:30 | 20.5686 N, 53.0940 E | 36.7       | 986           | 19.2   |
| Trip 5 (2025) | 49         | 14/10/2025 | 16:20 | 20.6447 N, 53.2879 E | 35.2       | 986           | 20.4   |
| Trip 5 (2025) | 50         | 14/10/2025 | 17:00 | 20.7139 N, 53.4123 E | 34.7       | 984           | 20.9   |
| Trip 5 (2025) | 51         | 15/10/2025 | 06:41 | 20.7987 N, 53.4870 E | 31.1       | 990           | 36.3   |
| Trip 5 (2025) | 52         | 15/10/2025 | 07:09 | 20.8278 N, 53.5784 E | 33.5       | 990           | 34.1   |
| Trip 5 (2025) | 53         | 15/10/2025 | 07:59 | 20.8515 N, 53.7579 E | 33.7       | 990           | 32.4   |
| Trip 5 (2025) | 54         | 15/10/2025 | 09:14 | 20.9202 N, 53.9622 E | 37.5       | 992           | 24.2   |
| Trip 5 (2025) | 55         | 15/10/2025 | 09:59 | 20.9023 N, 54.0024 E | 39.3       | 988           | 21.4   |
| Trip 5 (2025) | 56         | 15/10/2025 | 10:38 | 20.8216 N, 54.2239 E | 42.8       | 988           | 19.8   |
| Trip 5 (2025) | 57         | 15/10/2025 | 11:18 | 20.7350 N, 54.4176 E | 39.4       | 986           | 20.3   |
| Trip 5 (2025) | 58         | 15/10/2025 | 11:41 | 20.6654 N, 54.5933 E | 41.8       | 986           | 18.3   |
| Trip 5 (2025) | 59         | 15/10/2025 | 11:59 | 20.6563 N, 54.6359 E | 40.2       | 984           | 20.7   |
| Trip 5 (2025) | 60         | 15/10/2025 | 13:55 | 20.7064 N, 54.7321 E | 35.9       | 988           | 25.5   |
| Trip 5 (2025) | oil spill  | 12/10/2025 | 10:20 | 19.4054 N, 45.7271 E | 34.1       | 924           | 17.1   |
| Trip 5 (2025) | hot spring | 14/10/2025 | 07:58 | 20.3972 N, 50.9655 E | 32.7       | 984           | 22.8   |

*Metadata QC.* For Trip 2, the eight values 34.5–41.9 occur in the source humidity column while the temperature column is empty. The ledger assigns them to temperature; neither pressure nor relative humidity is inferred. Trip 3 site 21 is retained as 31.321% RH because

the original workbook records the fraction 0.31321. The 15 March records appended to the legacy Trip 3 worksheet are dated 2023 in that workbook; the curated table therefore labels them Trip 1 auxiliary/revisit records, and Trip 3 ends on 21 February 2024. The Trip 1 and Trip 3 Site 52 rows (and the appended 52AD record) repeat the Site 53 coordinates; the ledger restores the Site 52 location recorded in Trips 4 and 5 (20.8278°N, 53.5784°E), which the Trip 3 GPS track corroborates. Trip 5 site 40 contains the source value 194% RH, which is outside the physical range. Because no primary source currently establishes a decimal correction, the curated value is missing; 194 is preserved in the correction ledger.

**Table 3:** Audited XRF reporting-channel cross-references. Labels, formulas, charge and entity types were checked against pinned ChEBI and PubChem records.

| Analyte | Formula/type | ChEBI       | PubChem     |
|---------|--------------|-------------|-------------|
| Ag      | Ag           | CHEBI:30512 | CID:23954   |
| Al      | Al           | CHEBI:28984 | CID:5359268 |
| Al2O3   | Al2O3        | CHEBI:30187 | CID:9989226 |
| As      | As           | CHEBI:27563 | CID:5359596 |
| Au      | Au           | CHEBI:29287 | CID:23985   |
| Ba      | Ba           | CHEBI:32594 | CID:5355457 |
| BaO     | BaO          | –           | CID:62392   |
| Bi      | Bi           | CHEBI:33301 | CID:5359367 |
| Br      | Br           | CHEBI:22927 | CID:5360770 |
| Ca      | Ca           | CHEBI:22984 | CID:5460341 |
| CaO     | CaO          | CHEBI:31344 | CID:14778   |
| Cd      | Cd           | CHEBI:22977 | CID:23973   |
| Ce      | Ce           | CHEBI:33369 | CID:23974   |
| CeO2    | CeO2         | CHEBI:79089 | CID:73963   |
| Cl      | Cl           | CHEBI:23116 | CID:5360523 |
| Co      | Co           | CHEBI:27638 | CID:104730  |
| CoO     | CoO          | –           | CID:14786   |
| Cr      | Cr           | CHEBI:28073 | CID:23976   |
| Cr2O3   | Cr2O3        | CHEBI:48242 | CID:517277  |
| Cs      | Cs           | CHEBI:30514 | CID:5354618 |
| Cu      | Cu           | CHEBI:28694 | CID:23978   |
| CuO     | CuO          | CHEBI:75955 | CID:164827  |
| Dy      | Dy           | CHEBI:33377 | CID:23912   |
| Er      | Er           | CHEBI:33379 | CID:23980   |
| Eu      | Eu           | CHEBI:32999 | CID:23981   |
| Fe      | Fe           | CHEBI:18248 | CID:23925   |

| Analyte | Formula/type             | ChEBI       | PubChem     |
|---------|--------------------------|-------------|-------------|
| Fe2O3   | Fe2O3                    | CHEBI:50819 | CID:518696  |
| Ga      | Ga                       | CHEBI:49631 | CID:5360835 |
| Ga2O3   | Ga2O3                    | –           | CID:158605  |
| Gd      | Gd                       | CHEBI:33375 | CID:23982   |
| Ge      | Ge                       | CHEBI:30441 | CID:6326954 |
| Hf      | Hf                       | CHEBI:33343 | CID:23986   |
| Hg      | Hg                       | CHEBI:25195 | CID:23931   |
| Ho      | Ho                       | CHEBI:49648 | CID:23988   |
| I       | I                        | CHEBI:24859 | CID:5360629 |
| Ir      | Ir                       | CHEBI:49666 | CID:23924   |
| K       | K                        | CHEBI:26216 | CID:5462222 |
| K2O     | K2O                      | CHEBI:88321 | CID:9989219 |
| LE      | instrument_pseudoanalyte | –           | –           |
| La      | La                       | CHEBI:33336 | CID:23926   |
| La2O3   | La2O3                    | –           | CID:150906  |
| Lu      | Lu                       | CHEBI:33382 | CID:23929   |
| Mg      | Mg                       | CHEBI:25107 | CID:5462224 |
| MgO     | MgO                      | CHEBI:31794 | CID:14792   |
| Mn      | Mn                       | CHEBI:18291 | CID:23930   |
| MnO     | MnO                      | –           | CID:14940   |
| Mo      | Mo                       | CHEBI:28685 | CID:23932   |
| Na      | Na                       | CHEBI:26708 | CID:5360545 |
| Na2O    | Na2O                     | –           | CID:73971   |
| Nb      | Nb                       | CHEBI:33344 | CID:23936   |
| Nd      | Nd                       | CHEBI:33372 | CID:23934   |
| Nd2O3   | Nd2O3                    | –           | CID:159373  |
| Ni      | Ni                       | CHEBI:28112 | CID:935     |
| NiO     | NiO                      | –           | CID:14805   |

| Analyte | Formula/type | ChEBI       | PubChem     |
|---------|--------------|-------------|-------------|
| Os      | Os           | CHEBI:30687 | CID:23937   |
| P       | P            | CHEBI:28659 | CID:5462309 |
| P2O5    | P2O5         | CHEBI:37376 | CID:6326975 |
| Pb      | Pb           | CHEBI:25016 | CID:5352425 |
| PbO     | PbO          | CHEBI:81045 | CID:14827   |
| Pr      | Pr           | CHEBI:49828 | CID:23942   |
| PrO2    | PrO2         | —           | —           |
| Pt      | Pt           | CHEBI:33364 | CID:23939   |
| Rb      | Rb           | CHEBI:33322 | CID:5357696 |
| Re      | Re           | CHEBI:49882 | CID:23947   |
| S       | S            | CHEBI:26833 | CID:5362487 |
| SO3     | SO3          | CHEBI:29384 | CID:24682   |
| Sb      | Sb           | CHEBI:30513 | CID:5354495 |
| Sc      | Sc           | CHEBI:33330 | CID:23952   |
| Sc2O3   | Sc2O3        | —           | CID:134661  |
| Se      | Se           | CHEBI:27568 | CID:6326970 |
| Si      | Si           | CHEBI:27573 | CID:5461123 |
| SiO2    | SiO2         | CHEBI:30563 | CID:24261   |
| Sm      | Sm           | CHEBI:33374 | CID:23951   |
| Sm2O3   | Sm2O3        | —           | CID:159425  |
| Sn      | Sn           | CHEBI:27007 | CID:5352426 |
| Sr      | Sr           | CHEBI:33324 | CID:5359327 |
| SrO     | SrO          | —           | CID:73975   |
| Ta      | Ta           | CHEBI:33348 | CID:23956   |
| Tb      | Tb           | CHEBI:33376 | CID:23958   |
| Th      | Th           | CHEBI:33385 | CID:23960   |
| Ti      | Ti           | CHEBI:33341 | CID:23963   |
| TiO2    | TiO2         | CHEBI:32234 | CID:26042   |

| Analyte | Formula/type | ChEBI       | PubChem     |
|---------|--------------|-------------|-------------|
| Tm      | Tm           | CHEBI:33380 | CID:23961   |
| U       | U            | CHEBI:27214 | CID:23989   |
| V       | V            | CHEBI:27698 | CID:23990   |
| V2O5    | V2O5         | CHEBI:30045 | CID:14814   |
| W       | W            | CHEBI:27998 | CID:23964   |
| Y       | Y            | CHEBI:33331 | CID:23993   |
| Yb      | Yb           | CHEBI:33381 | CID:23992   |
| Zn      | Zn           | CHEBI:27363 | CID:23994   |
| ZnO     | ZnO          | CHEBI:36560 | CID:3007857 |
| Zr      | Zr           | CHEBI:33342 | CID:23995   |
| ZrO2    | ZrO2         | —           | CID:62395   |

*Interpretation.* A dash denotes an audited null, not a missing review. These `rdfs:seeAlso` cross-references identify the element or oxide-equivalent reporting channel; they do not assert that the corresponding molecule or mineral occurs in a sample. The  $\text{P}_2\text{O}_5$  channel is cross-referenced to the ChEBI  $\text{P}_4\text{O}_{10}$  class only because the formulas share the P:O empirical ratio. LE is an instrument aggregate, and  $\text{PrO}_2$  has no formula-matching external identifier in the audited sources.

## References

- [1] Degtyarenko, K., Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcántara, R., Darsow, M., Guedj, M., Ashburner, M.: ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36**(suppl\_1), 344–350 (2008) <https://doi.org/10.1093/nar/gkm791>
- [2] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., *et al.*: PubChem 2023 update. *Nucleic Acids Research* **51**(D1), 1373–1380 (2023)